THE TWO-FOLD DERIVED BASE CHANGE COMPARISON FOR A COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- K-flat complex, defined by the property that tensoring with it preserves quasi-isomorphisms. definition at source lines 14185-14191 of the same chapter. Supplies the kind of resolution used in the proof.
- Derived tensor product may be computed with a K-flat representative. used at proof lines 14821-14860. Justifies replacing the derived constructions by ordinary ones.
- Associativity of the total tensor product of complexes. used at proof lines 14860-14891. Performs the final identification of the two sides.

Source reference. Original lemma 08YU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11352–11352. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 06Y4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8873–8873. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BYK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15018–15018. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06Y1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8870–8870. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0795: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9260–9260. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 08BI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10568–10568. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06Y6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8875–8875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BYI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15016–15016. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BYJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15017–15017. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B \to C$ be ring maps. Let $N^\bullet$ be a complex of B-modules and $K^\bullet$ a complex of C-modules. The compositions of the functors*

$$D(A) \xrightarrow{-\otimes^{\mathbf{L}}_A N^\bullet} D(B) \xrightarrow{-\otimes^{\mathbf{L}}_B K^\bullet} D(C)$$

is the functor $- \otimes^{\mathbf{L}}_A (N^\bullet \otimes^{\mathbf{L}}_B K^\bullet) : D(A) \to D(C)$. If M, N, K are modules over A, B, C, then we have

$$(M \otimes^{\mathbf{L}}_A N) \otimes^{\mathbf{L}}_B K = M \otimes^{\mathbf{L}}_A (N \otimes^{\mathbf{L}}_B K) = (M \otimes^{\mathbf{L}}_A C) \otimes^{\mathbf{L}}_C (N \otimes^{\mathbf{L}}_B K)$$

in $D(C)$. We also have a canonical isomorphism

$$(M \otimes^{\mathbf{L}}_A N) \otimes^{\mathbf{L}}_B K \longrightarrow (M \otimes^{\mathbf{L}}_A K) \otimes^{\mathbf{L}}_C (N \otimes^{\mathbf{L}}_B C)$$

using signs. Similar results holds for complexes.

Proof. Choose a K-flat complex $P^\bullet$ of B-modules and a quasi-isomorphism $P^\bullet \to N^\bullet$ (Lemma 06Y4). Let $M^\bullet$ be a K-flat complex of A-modules representing an arbitrary object of $D(A)$. Then we see that

$$(M^\bullet \otimes^{\mathbf{L}}_A P^\bullet) \otimes^{\mathbf{L}}_B K^\bullet \longrightarrow (M^\bullet \otimes^{\mathbf{L}}_A N^\bullet) \otimes^{\mathbf{L}}_B K^\bullet$$

is an isomorphism by Lemma 0BYK applied to the material inside the brackets. By Lemmas 06Y1 and 0795 the complex

$$\mathrm{Tot}(M^\bullet \otimes_A P^\bullet) = \mathrm{Tot}((M^\bullet \otimes_R A) \otimes_A P^\bullet$$

is K-flat as a complex of B-modules and it represents the derived tensor product in $D(B)$ by construction. Hence we see that $(M^\bullet \otimes^{\mathbf{L}}_A P^\bullet) \otimes^{\mathbf{L}}_B K^\bullet$ is represented by the complex

$$\mathrm{Tot}(\mathrm{Tot}(M^\bullet \otimes_A P^\bullet) \otimes_B K^\bullet) = \mathrm{Tot}(M^\bullet \otimes_A \mathrm{Tot}(P^\bullet \otimes_B K^\bullet))$$

of C-modules. Equality by Homology, Remark 08BI. Going back the way we came we see that this is equal to

$$M^\bullet \otimes^{\mathbf{L}}_A (P^\bullet \otimes^{\mathbf{L}}_B K^\bullet) \longleftarrow M^\bullet \otimes^{\mathbf{L}}_A (N^\bullet \otimes^{\mathbf{L}}_B K^\bullet)$$

The arrow is an isomorphism by definition of the functor $- \otimes^{\mathbf{L}}_B K^\bullet$. All of these constructions are functorial in the complex $M^\bullet$ and hence we obtain our isomorphism of functors.

By the above we have the first equality in

$$(M \otimes^{\mathbf{L}}_A N) \otimes^{\mathbf{L}}_B K = M \otimes^{\mathbf{L}}_A (N \otimes^{\mathbf{L}}_B K) = (M \otimes^{\mathbf{L}}_A C) \otimes^{\mathbf{L}}_C (N \otimes^{\mathbf{L}}_B K)$$

The second equality follows from the final statement of Lemma 06Y6. The same thing allows us to write $N \otimes^{\mathbf{L}}_B K = (N \otimes^{\mathbf{L}}_B C) \otimes^{\mathbf{L}}_C K$ and substituting we get

$$\begin{aligned}
(M \otimes^{\mathbf{L}}_A N) \otimes^{\mathbf{L}}_B K &= (M \otimes^{\mathbf{L}}_A C) \otimes^{\mathbf{L}}_C ((N \otimes^{\mathbf{L}}_B C) \otimes^{\mathbf{L}}_C K) \\
&= (M \otimes^{\mathbf{L}}_A C) \otimes^{\mathbf{L}}_C (K \otimes^{\mathbf{L}}_C (N \otimes^{\mathbf{L}}_B C)) \\
&= ((M \otimes^{\mathbf{L}}_A C) \otimes^{\mathbf{L}}_C K) \otimes^{\mathbf{L}}_C (N \otimes^{\mathbf{L}}_B C)) \\
&= (M \otimes^{\mathbf{L}}_C K) \otimes^{\mathbf{L}}_C (N \otimes^{\mathbf{L}}_B C)
\end{aligned}$$

by Lemmas 0BYI and 0BYJ as well as the previously mentioned lemma. □